Properties of the Discrete Fourier Transform

Exercise 2 numerical verification and recursive FFT report

This report checks the algebraic properties of the DFT on a reproducible real-valued test signal of length N=32. All calculations use the convention:

$$A_k = \sum_{j=0}^{N-1} a_j e^{-2\pi i k j / N}$$

Executive summary

Property	Maximum error	Tolerance	Status
Conjugate symmetry	8.406e-14	1e-10	PASS
Circular shift theorem	8.632e-14	1e-10	PASS
Parseval theorem	7.105e-15	1e-10	PASS
Cooley--Tukey recombination	5.844e-14	1e-10	PASS
Recursive radix-2 FFT	3.972e-15	1e-10	PASS

Conjugate symmetry. For a real signal, the coefficient at k equals the complex conjugate of the coefficient at (–k) modulo N.

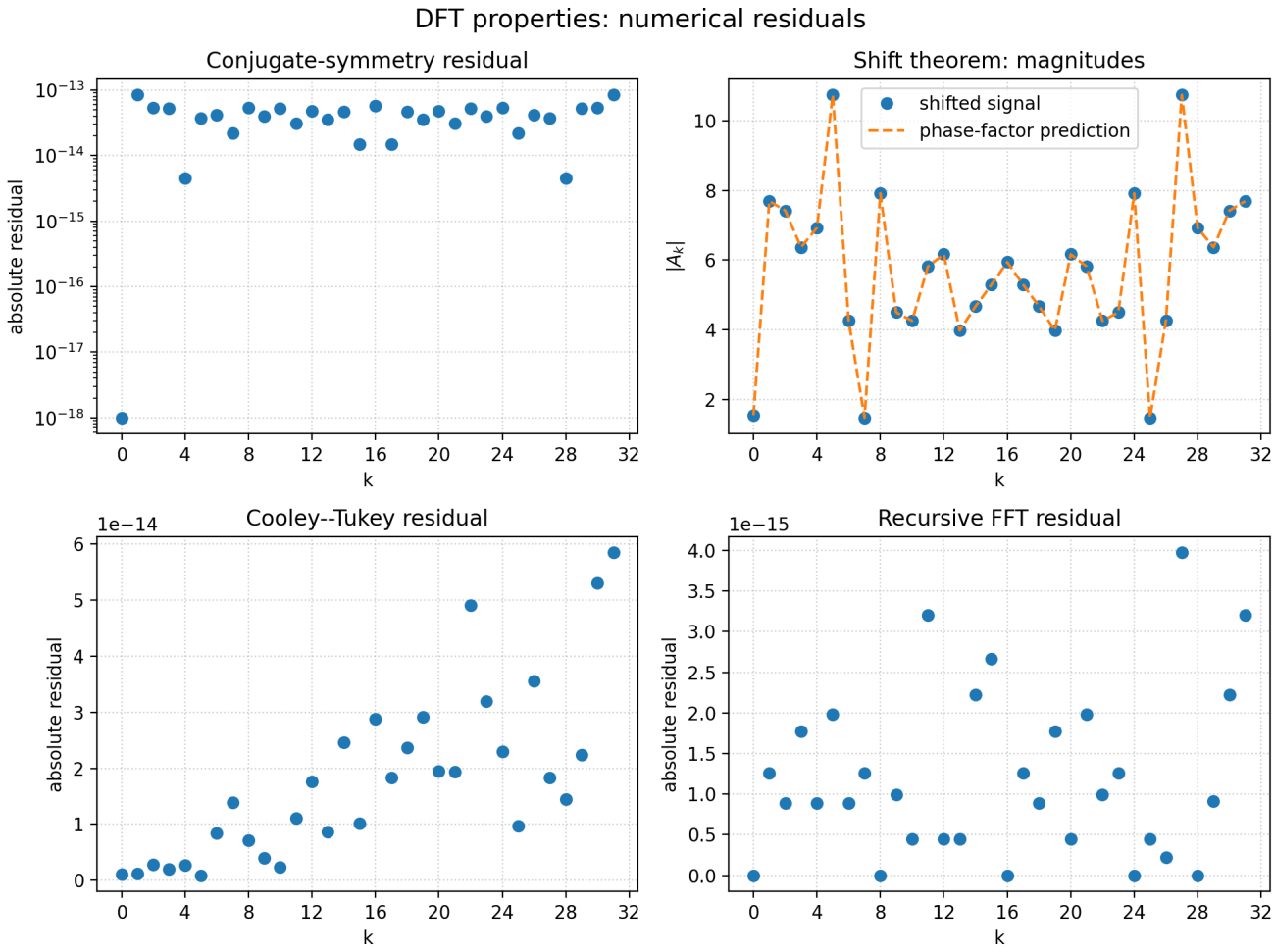
Circular shift theorem. Rolling the samples by $m=7$ multiplies A_k by $e^{-2\pi i k m/N}$.

Parseval theorem. The sum of |a■|² equals 1/N times the sum of |A■|² for this DFT convention.

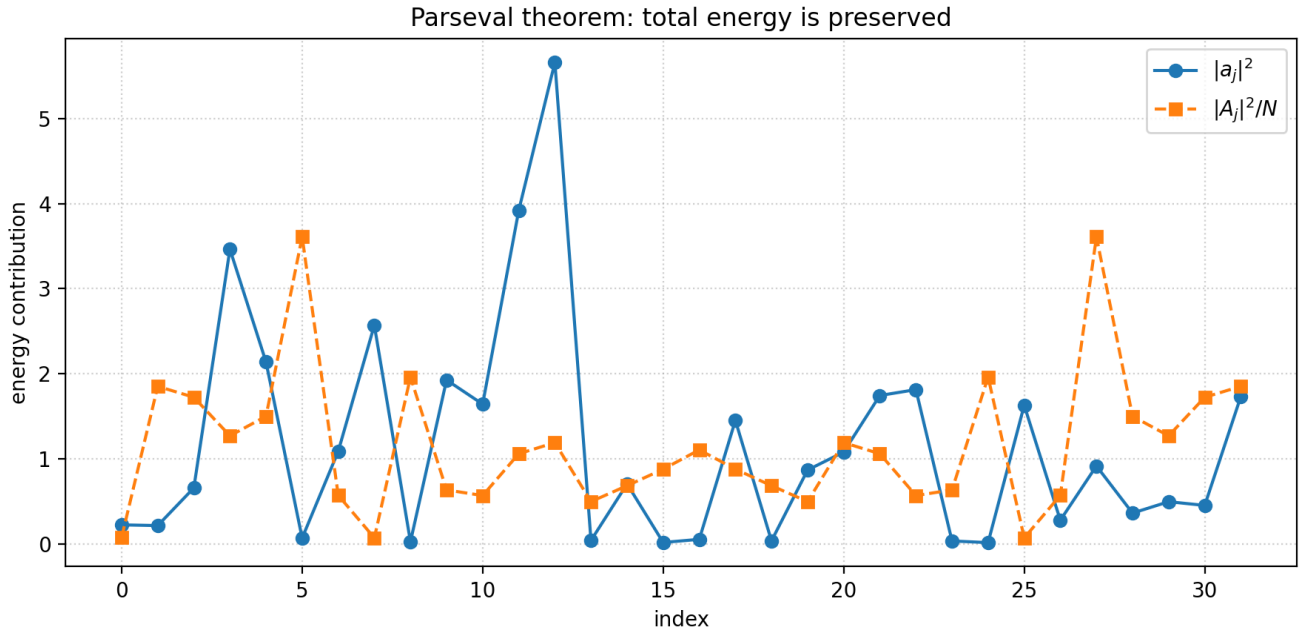
Cooley--Tukey recombination. The even and odd transforms combine as A■ = E■ + ω■■O■ and A■■■/■ = E■ – ω■■O■.

Recursive radix-2 FFT. The recursive implementation agrees with NumPy's FFT.

Results and visual checks



The residuals are at floating-point round-off level. The shift theorem changes phase but not spectral magnitude, which is why the two magnitude curves overlap.



Parseval check: signal energy = 37.27627450, coefficient energy / N = 37.27627450.

Cooley--Tukey recursion and radix-2 FFT

Splitting the samples into even and odd subsequences gives E■ = DFT(a■, a■, …) and O■ = DFT(a■, a■, …). With ω■ = exp(–2πi/N), applying the identities below recursively produces the radix-2 FFT, reducing the operation count from O(N²) to O(N log■ N).

$$A_k = E_k + \omega_N^k O_k, \qquad A_{k+N/2} = E_k - \omega_N^k O_k$$

N	Recursive FFT [s]	NumPy FFT [s]	Ratio
8	4.230e-05	4.300e-06	9.84
16	9.030e-05	4.100e-06	22.02
32	1.828e-04	4.100e-06	44.59
64	3.740e-04	4.300e-06	86.98
128	7.543e-04	5.100e-06	147.90
256	1.540e-03	8.900e-06	173.01
512	3.019e-03	9.300e-06	324.61
1024	6.058e-03	1.380e-05	438.96

